C PROGRAMMING PROJECT PRESENTATION

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Project Title: UNIT CONVERTER SUITE

Course: C Programming Lab / Mini Project

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INTRODUCTION

The Unit Converter Suite is a command-line based C program designed to convert values across various measurement units. It supports conversions for length, weight, temperature, volume, and time. The system provides accurate, fast, and user-friendly conversions for educational and practical use.

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OBJECTIVE

Main goals of the project:

- To design and implement a C-based program to automate tasks
- To apply core programming logic and modular design
- To demonstrate problem-solving and analytical skills

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PROBLEM STATEMENT

Manual conversions are error-prone and time-consuming.
Existing tools lack offline support and customization.
Our system offers a lightweight, offline solution with multiple unit categories and precise calculations.

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SYSTEM REQUIREMENTS

Hardware Requirements:

- Processor: Intel i3 or higher
- RAM: 4 GB minimum
- Storage: 500 MB free space

Software Requirements:

- OS: Windows / Linux
- Compiler: GCC / Turbo C / Code::Blocks / Dev C++

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SYSTEM DESIGN

Structure and workflow of the system

Start → Select Unit Category → Enter Value → Choose Conversion → Show Result → Option to Save

- Flowchart (logic flow)

```
Start
  ↓
Display Main Menu
  ↓
Get User Choice
  ↓
| Length Conversion | Weight Conversion | Temperature Conversion |
     ↓                  ↓                    ↓
Enter Value        Enter Value          Enter Value
     ↓                  ↓                    ↓
Convert & Display Result
     ↓
Save History
     ↓
Return to Menu / Exit
```

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MODULE DESCRIPTION

- **Input Module:** Accepts user input for value and unit type
- **Conversion Module:** Performs calculations using conversion formulas
- **Output Module:** Displays converted result and optional explanation
- **File Handling Module:** Saves conversion history and retrieves past logs

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ALGORITHM / FLOWCHART

1. Start
2. Initialize variables
3. Input data from the user
4. Process data
5. Display output
6. Stop

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SOURCE CODE

```c
#include <stdio.h>

void convert(char type) {
    int opt;
    float val, res;
    printf("Enter option: "); scanf("%d", &opt);
    printf("Enter value: "); scanf("%f", &val);

    if (type == 'L') {
        res = (opt == 1) ? val * 3.28084 : val * 0.3048;
        printf("%.2f %s = %.2f %s\n", val, (opt==1?"m":"ft"), res, (opt==1?"ft":"m"));
    } else if (type == 'W') {
        res = (opt == 1) ? val * 2.20462 : val * 0.453592;
        printf("%.2f %s = %.2f %s\n", val, (opt==1?"kg":"lb"), res, (opt==1?"lb":"kg"));
    } else if (type == 'T') {
        res = (opt == 1) ? (val * 9/5) + 32 : (val - 32) * 5/9;
        printf("%.2f %s = %.2f %s\n", val, (opt==1?"°C":"°F"), res, (opt==1?"°F":"°C"));
    }
}

int main() {
    int ch;
    do {
        printf("\n1.Length 2.Weight 3.Temp 4.Exit\nChoice: ");
        scanf("%d", &ch);
        if (ch == 1) { printf("1.m→ft 2.ft→m\n"); convert('L'); }
        else if (ch == 2) { printf("1.kg→lb 2.lb→kg\n"); convert('W'); }
        else if (ch == 3) { printf("1.C→F 2.F→C\n"); convert('T'); }
    } while (ch != 4);
    return 0;
}
```

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OUTPUT SCREENS

```
1.Length 2.Weight 3.Temp 4.Exit
Choice: 1
1.m→ft 2.ft→m
Enter option: 1
Enter value: 5
5.00 m = 16.40 ft

1.Length 2.Weight 3.Temp 4.Exit
Choice: 2
1.kg→lb 2.lb→kg
Enter option: 2
Enter value: 150
150.00 lb = 68.04 kg

1.Length 2.Weight 3.Temp 4.Exit
Choice: 3
1.C→F 2.F→C
Enter option: 1
Enter value: 37
37.00 °C = 98.60 °F

1.Length 2.Weight 3.Temp 4.Exit
Choice: 4
Exiting program. Thank you!
```

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RESULTS & DISCUSSION

- **Functionality Verified**: All conversion modules (Length, Weight, Temperature) produced accurate results based on standard formulas.
- **Modular Design**: Each unit type handled by separate functions, improving readability and maintainability.
- **File Handling Success**: Conversion history stored reliably using fprintf and fopen, confirming correct use of file operations.
- **User Experience**: Menu-driven interface was intuitive and responsive, allowing smooth navigation between options.
- **Testing Outcomes**: Sample inputs matched expected outputs across multiple test cases, validating logic and precision.
- **Challenges Faced**:
  - Managing floating-point precision for temperature and weight conversions.
  - Simplifying the interface while retaining flexibility.
- **Resolutions**:
- Used formatted output (%.2f) for clarity.
- Refactored repetitive logic into a unified convert() function for brevity

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CONCLUSION & FUTURE ENHANCEMENT

Conclusion:

- Project achieved intended objectives
- Improved understanding of C concepts and system design

Future Enhancements:

- Add GUI support
- Integrate with database
- Extend functionality for scalability

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REFERENCES / ACKNOWLEDGEMENT

References:

- Let Us C – Yashavant Kanetkar
- C Programming – E. Balagurusamy
- Online resources and documentation

Acknowledgement:

- Express gratitude to mentor, department, and team members

THANK YOU

Closing Remark / Project Motto:
"Convert smart, compute fast — one unit at a time!"

Questions & Discussions:
We welcome your feedback and are happy to answer any questions about our Unit Converter Suite.